Epistemic Network Models for the Misinformation Age



Greg Conradi Smith
Biomathematics Journal Club & Seminar
2022 September 16

W&M biomath

How does information move in groups of individuals and institutions?

How does that dynamic influence beliefs and actions?

Such questions have been addressed by philosophers of science, complex systems theorists, sociologists, political scientists, social psychologists, and marketing and management researchers.

Singer et al. 2021 (our reading) is a book chapter written by political philosophers.

They say that the notion of an **epistemic network model** has been helpful for understanding how individuals' beliefs can influence the beliefs of others.

We at the Biomath Seminar know a lot about networks and models!

What does epistemic (epistemology) mean?

Epistemic means relating to knowledge or to the degree of its validation.

We will look at some research articles that explore various epistemic network models and decide for ourselves … do they appear to be helpful?

But first, how did I get into this?

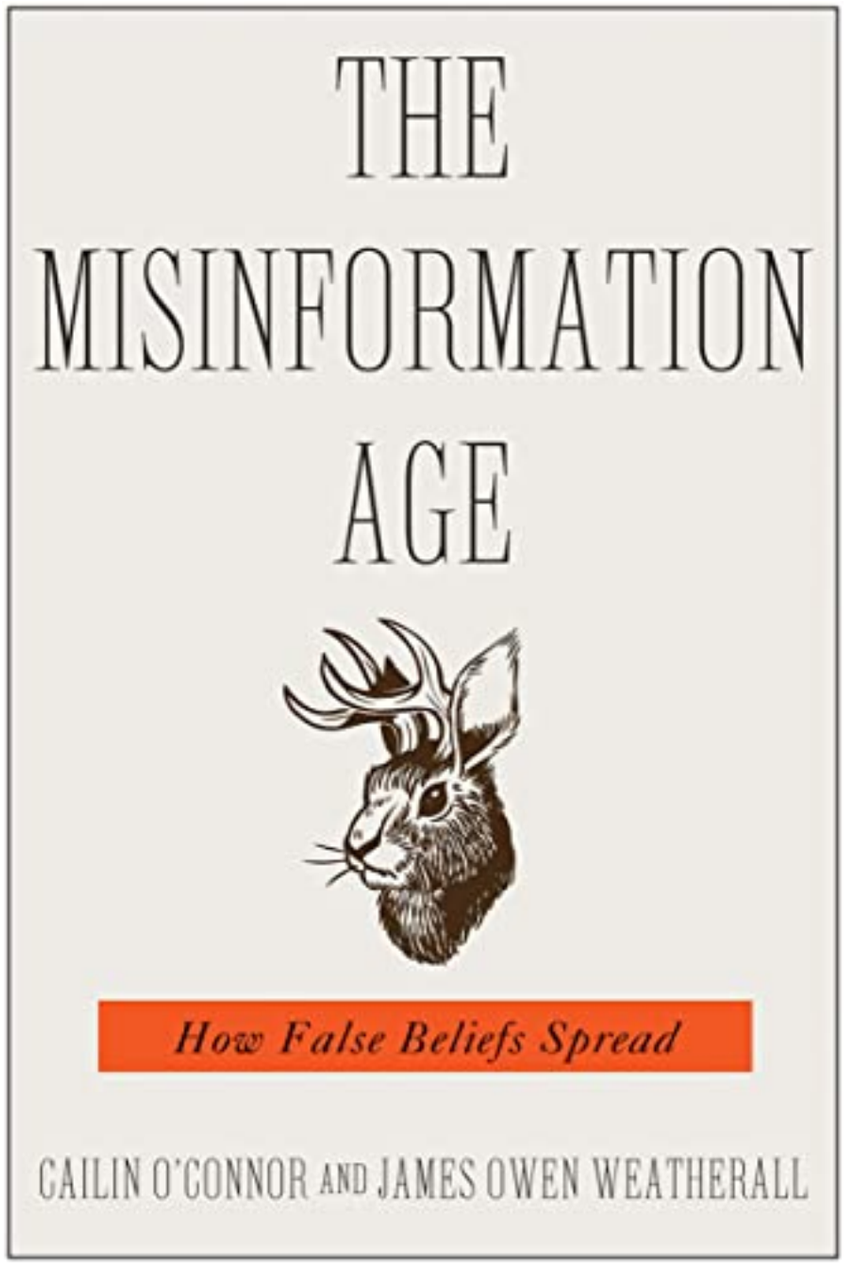
Singer, D. J., Grim, P., Bramson, A., Holman, B., Jung, J., & Berger, W. J. (2021). **Epistemic networks and polarization**. In:*The Routledge Handbook of Political Epistemology*.

Department of Logic and Philosophy of Science and Institute for Mathematical Behavioral Science at UC Irvine



Cailin O'Connor

I'm a philosopher of biology and behavioral sciences and evolutionary game theorist.



Yale University Press (2020)



James Owen Weatherall

I'm a physicist, philosopher, and mathematician.

Thesis: False beliefs persist and spread due to social factors - not just individual psychology

Cailin O'Connor (UC Irving) gives public lecture at Minnesota Center for the Philosophy of Science on September 19, 2019



https://youtu.be/qSHEkZP_KRY?t=254

start at 4:13

Example: Spread of gossip in a social network

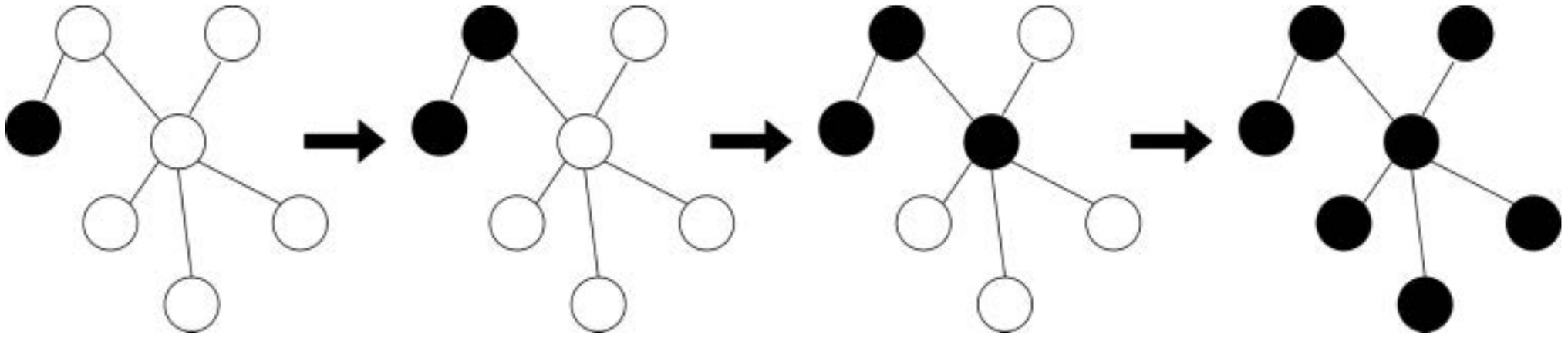


Figure 12.2 Illustration of a high school gossip epistemic network through time.

On the left, we see Charlie, who uses a piece of (mis)information to start the rumor. In the next step, that information has spread to Charlie's friend. In the time step after that, the misinformation is spread to the popular student, who shares it with the rest of the group. Here the color of the agent represents whether the rumor has spread to that agent.

Analogous to a **contagion model** of a pathogen spreading through social contacts.
Each individual can adopt a set of states, e.g., **susceptible** and **infected**.

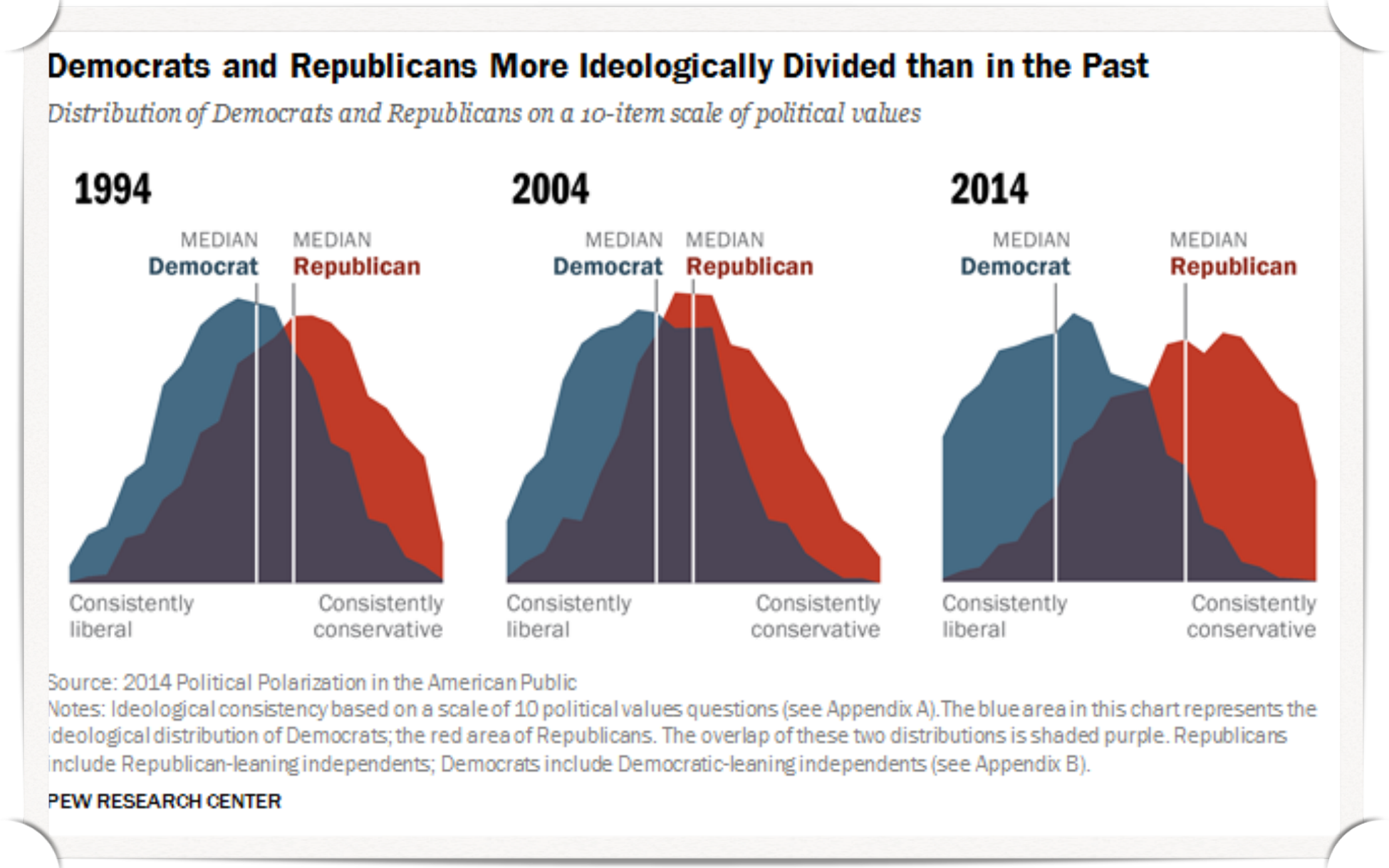


Is epistemic network modeling a trivial recasting of epidemiological modeling?

Our focus today will be **polarization** in epistemic networks, because this dynamic seemed likely be be more than recasting of epidemiological ideas.

By polarization we mean **persistent disagreement between groups**, often two, perhaps across an array of different issues regarding belief or opinion.

For example, in many **democratic** countries the **political climate is polarized**.



Long before computational modeling of epistemic networks was possible, **empirical work** showed that discussion tends to lead to greater agreement among individuals.

In a classic postwar study, Festinger et al. (1950) showed how **location around housing courts**, and thus social interaction, **determined opinions** in a study of MIT students.



Figure 1. MIT Westgate housing for students, c. 1946 (Leon Festinger Papers, Bentley Library, University of Michigan).

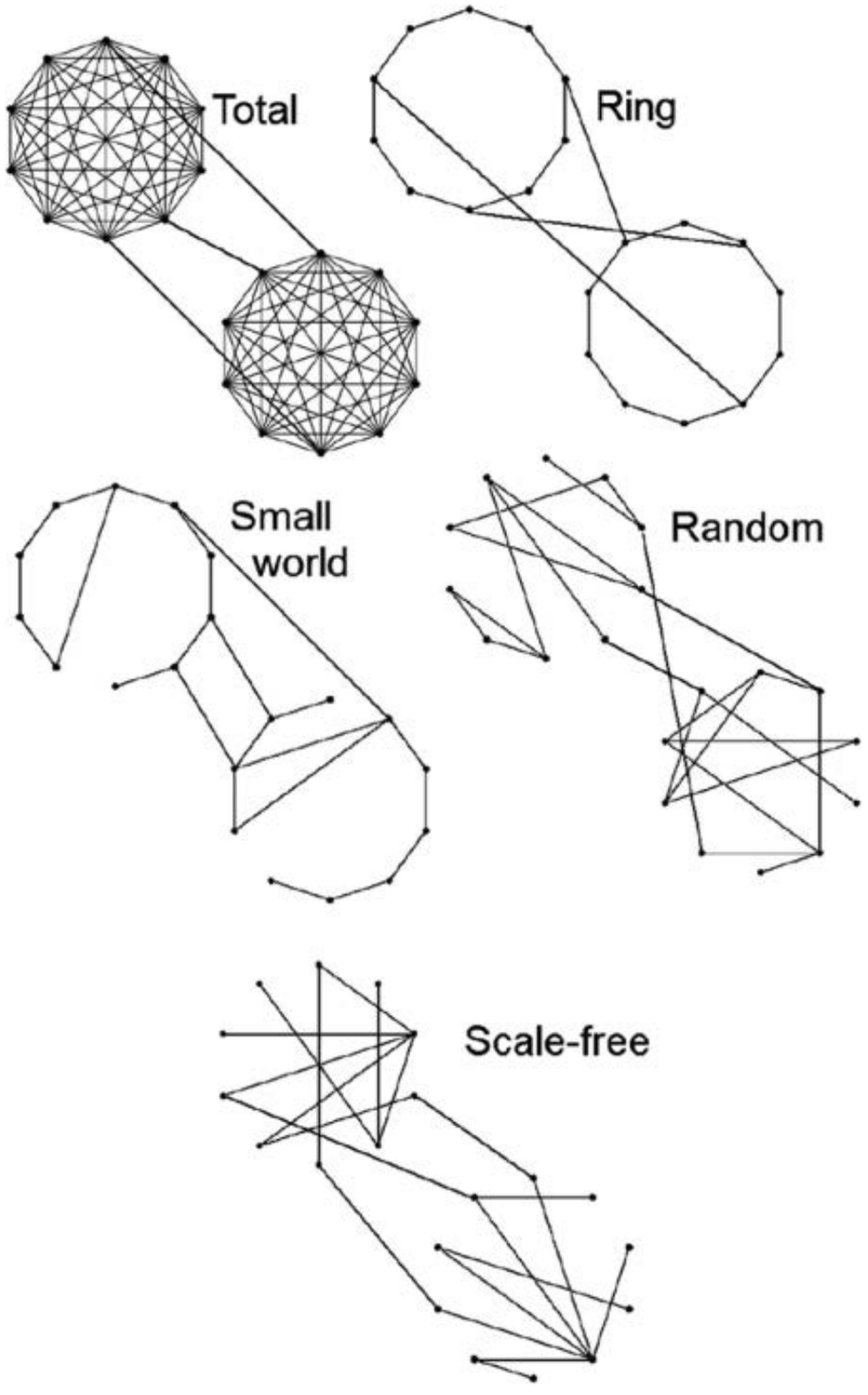
Students tended to adopt the beliefs of their neighbors.

These housing studies were a step toward social-science research on the effects of racial and class mixing in neighborhoods.

Studies of racially integrated housing argued for residential diversity and influenced the landmark case of Brown v. Board of Education.

Festinger, L., Schachter, S., Back, K. (1950). Social pressures in informal groups; a study of human factors in housing.
O'Connor, C., & Weatherall, J. O. (2018). Scientific polarization. *European Journal for Philosophy of Science 8*(3), 855-875.
Borrman, K. (2022). Studying Friendship in Housing the MIT School of Architecture at MIT in the Postwar Years. *Journal of Urban History*, *48*(5), 1100-1120.

What about **computational models** of opinion dynamics?



A **static divided epistemic network** is a possible starting point for modeling polarization of belief or opinion.

Network edges represent connections within and between different social groups.

Under the assumption that interaction leads to greater agreement among individuals

The network dynamics usually destroy whatever polarization in opinion was originally present (Singer et al. 2021).

There are many computational models of social influence that attempt to capture and explain this tendency towards shared opinions.

The question for those interested in polarization is why, despite this general tendency, some groups never converge to uniform beliefs.

What conditions lead to loss of consensus and development of polarized opinion?

The epistemic network modeling literature shows that a key ingredient is to **make the social influence of one individual on another dependent on the similarity of their beliefs**.

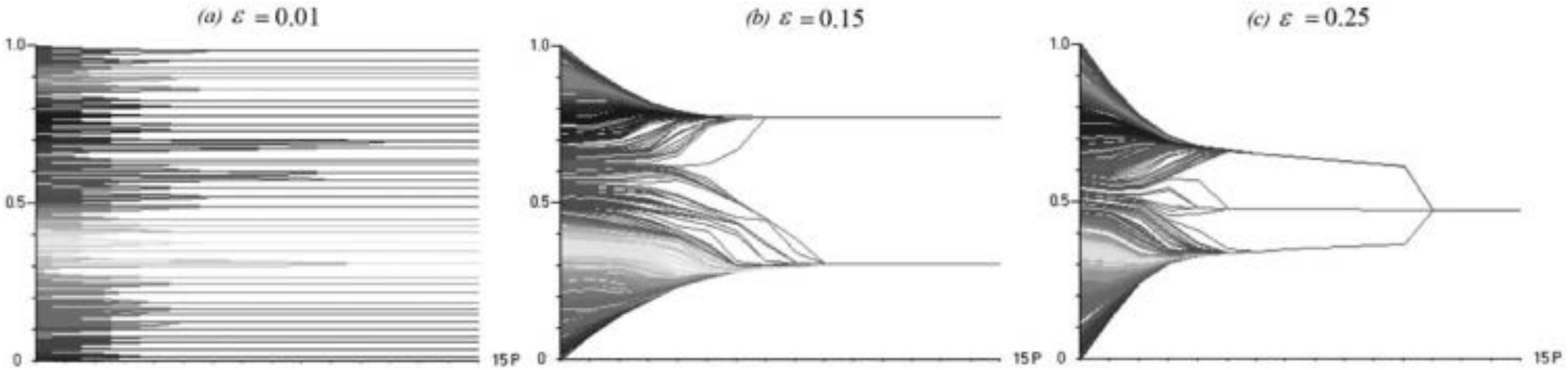
A variety of such "bounded confidence" models have been developed that show polarization of belief might arise in this way

A very simple example follows.

bounded confidence model

This model generates polarization with agents who update using others' opinions only when those opinions are close enough to their own. As each agent's beliefs change, so does the identity of the agents considered close enough.

Because `who listens to whom' changes this is, in effect, an **adaptive** network.



suspicious ⟷ trusting

Opinions in the model are mapped onto the [0, 1] interval with uniformly distributed initial opinions. Belief updating involves a weighted average of neighboring opinions.

Hegselmann & Krause 2002
see also Deffuant et al. 2002

European Journal for Philosophy of Science (2018) 8: 855–875
https://doi.org/10.1007/s13194-018-0213-9

PAPER IN GENERAL PHILOSOPHY OF SCIENCE

Scientific polarization

Cailin O'Connor[1] · **James Owen Weatherall**[1]

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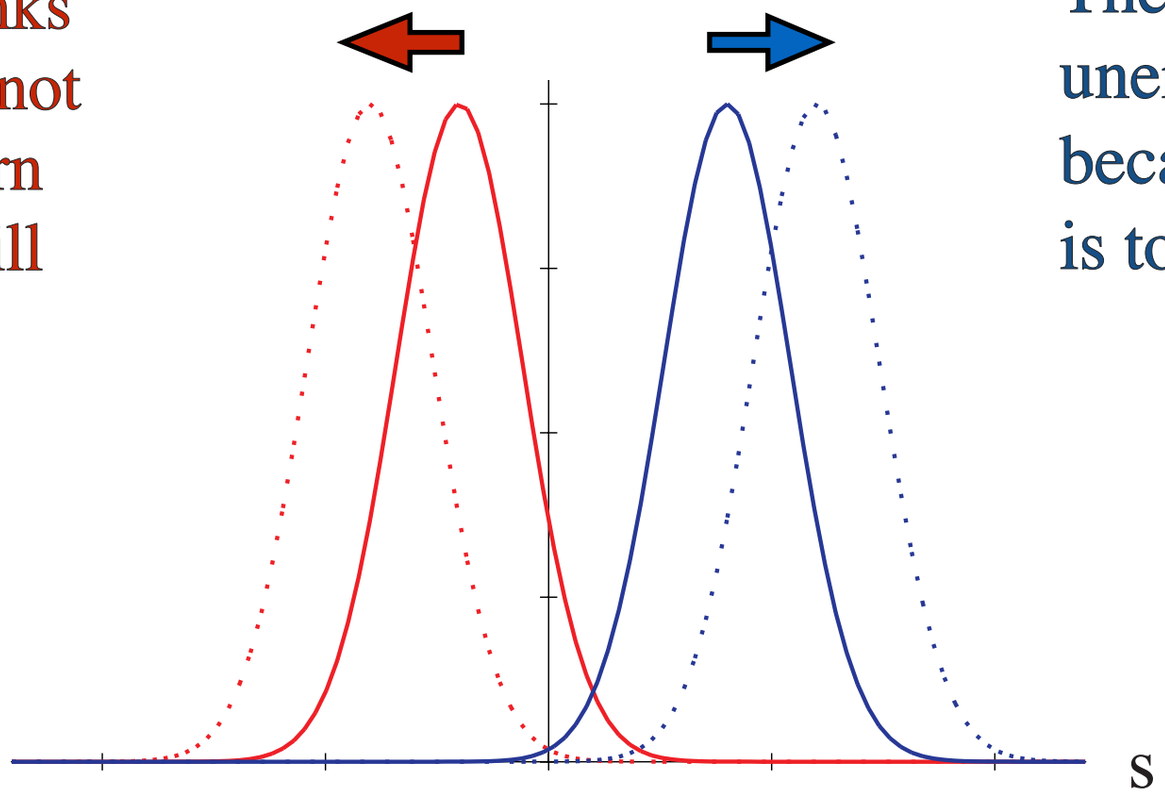
Abstract

Contemporary societies are often "polarized", in the sense that sub-groups within these societies hold stably opposing beliefs, even when there is a fact of the matter. Extant models of polarization do not capture the idea that some beliefs are true and others false. Here we present a model, based on the network epistemology framework of Bala and Goyal (Learning from neighbors, *Rev. Econ. Stud. 65*(3), 784–811 1998), in which polarization emerges even though agents gather evidence about their beliefs, and true belief yields a pay-off advantage. As we discuss, these results are especially relevant to polarization in scientific communities, for these reasons. The key mechanism that generates polarization involves treating evidence generated by other agents as uncertain when their beliefs are relatively different from one's own.

Even in the absence of social influence, belief polarization is not *necessarily* evidence of irrational behavior.

Two agents with different background beliefs (priors) may respond **rationally to the same evidence in the opposite direction**.

The Monetarist thinks that businesses are not hiring out of concern that loose policy will lead to inflation



The Keynesian think that unemployment is high because monetary policy is too tight

Polarization may occur without selective perception or confirmation bias.

But usually the difference between the posteriors (eventually) goes to zero.

Dixit, A.K., & Weibull, J.W. (2007). Political polarization. *Proceedings of the National Academy of Sciences*, *104*(18), 7351–7356.
Jern, A., Chang, K.-M.K., Kemp, C. (2014). Belief polarization is not always irrational. *Psychological Review*, *121*(2), 206.
Benoıt, J.-P., & Dubra, J. (2014). A theory of rational attitude polarization.

This next example model includes discussion of multiple issues, differing levels of interest, and opinion that change in response to interactions with others who have similar interests and opinions.

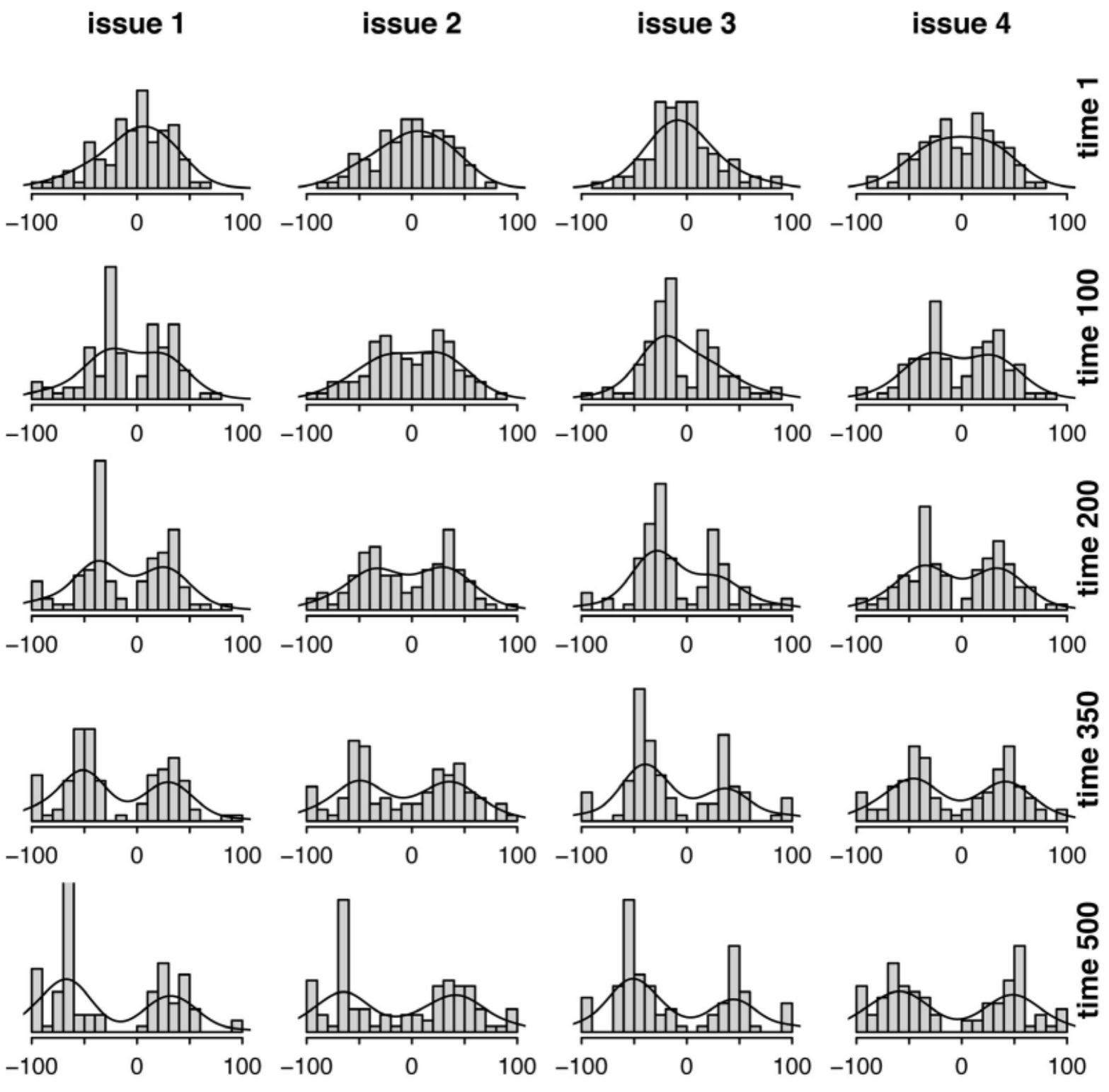
Both **polarization of beliefs** and **homophily**—where two groups stop interacting with each other based on their different beliefs — are observed.

Table 1. Outline of the Simulation Algorithm

Initial Conditions:
100 actors 4 issues; issue interest ~ normal ($\mu = 0$, $\sigma = 33.3$); interest range (–100; +100) Initialize perceived ideological distance λ = mean Euclidean distance among actors
Iteration Flow:
Selection of interaction partners At each iteration for each actor: Random sample of potential interlocutors ~ to the overall level of interest Draw from the sample the actual interlocutors with $p = 1 - \lambda$
Process of interpersonal influence For each pair of actors previously selected: Select the issue for discussion Compute the change for each actor based on their interest on the issue Determine the direction of change according to the sign of the issue Update actors' level of interest Update actors' perceived ideological distance with the current/actual distance Save all necessary information (Repeat 500 times)

Baldassarri and Bearman 2007

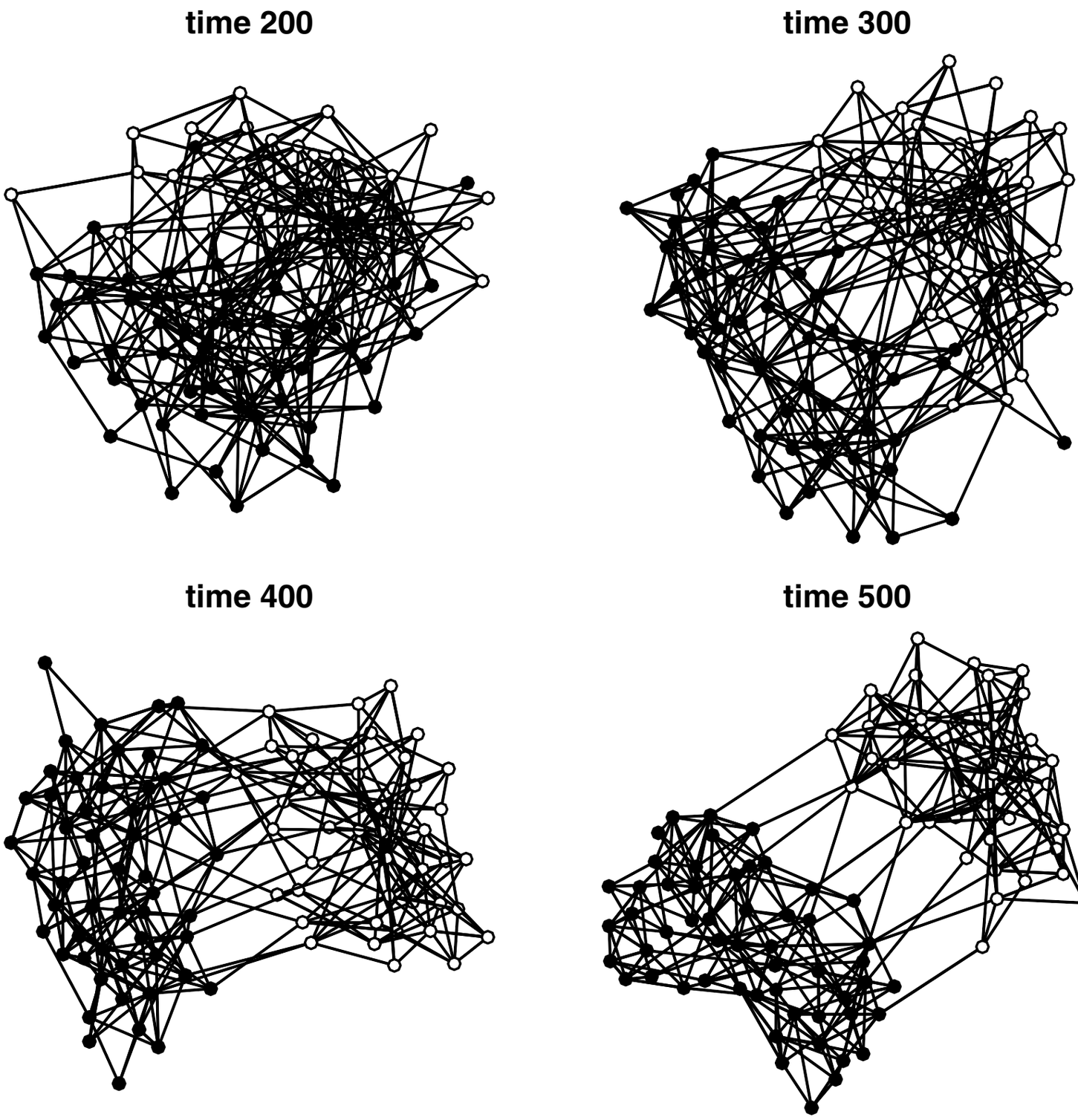
attitude distributions may polarize over time



Over time the number of actors with a modest commitment to issues decreases.

Attitudes become stronger, either positive or negative.

Both dispersion and bimodality increase.



Ideological polarization can occur on a single issue at a time.

A disproportionately discussed "takeoff issue" may lead to social structural change: polarized groups arise within which actors experience increasing homogeneity.

The opinions they hold on other issues may remain heterogeneous and/or moderate.

Figure 6. Discussion Network Over Time in a Takeoff Case (#963)

Notes: Actors (circles) are connected (by arcs) if their frequency of interaction is higher than expected by chance. Nodes are colored according to the sign of the most popular issue. Black is positive, white is negative. Simulation #963, takeoff.

Perhaps of interest: There is a potential mismatch between the perceived and actual level of opinion polarization across social categories

People will talk about trivial matters with most anyone, but for topics that are important to them, they disproportionately reveal their views to people who they believe share views broadly similar to their own.

Because actors tend to talk to others who share their beliefs, the discussion of important issues induces the experience of a segregated discussion network.

If no break in network symmetry - actors tend to perceive that their environments are polarized, even though this is not the case.

If there is a break in network symmetry - when a single issue dominates discussion, individuals overestimate ideological homogeneity because their contacts are disproportionately within a polarized subgroup.

In both cases, personal experience is misleading with respect to issue heterogeneity, because public opinion remains heterogeneous on the broad range of other issues.

Synthese (2021) 198 (Suppl 25):S6179–S6200
https://doi.org/10.1007/s11229-020-02675-3

S.I.: DISAGREEMENT IN SCIENCE

Endogenous epistemic factionalization

James Owen Weatherall[1] · Cailin O'Connor[1]

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© Springer Nature B.V. 2020

Abstract

Why do people who disagree about one subject tend to disagree about other subjects as well? In this paper, we introduce a model to explore this phenomenon of 'epistemic factionization'. Agents attempt to discover the truth about multiple propositions by testing the world and sharing evidence gathered. But agents tend to mistrust evidence shared by those who do not hold similar beliefs. This mistrust leads to the endogenous emergence of factions of agents with multiple, highly correlated, polarized beliefs.

Conformity has a negative effect on rational actors reaching consensus

Synthese (2021) 198:7257–7278
https://doi.org/10.1007/s11229-019-02520-2

Conformity in scientific networks

James Owen Weatherall[1] · Cailin O'Connor[1]

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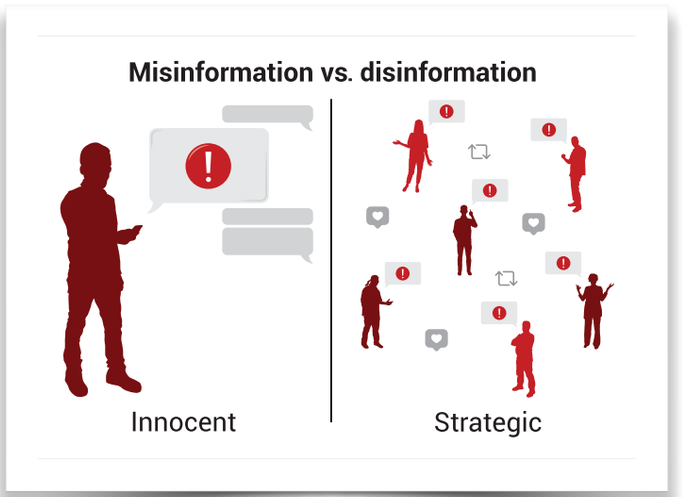
Abstract
Scientists are generally subject to social pressures, including pressures to conform with others in their communities, that affect achievement of their epistemic goals. Here we analyze a network epistemology model in which agents, all else being equal, prefer to take actions that conform with those of their neighbors. This preference for conformity interacts with the agents' beliefs about which of two (or more) possible actions yields the better result. We find a range of possible outcomes, including stable polarization in belief and action. The model results are sensitive to network structure. In general, though, conformity has a negative effect on a community's ability to reach accurate consensus about the world.

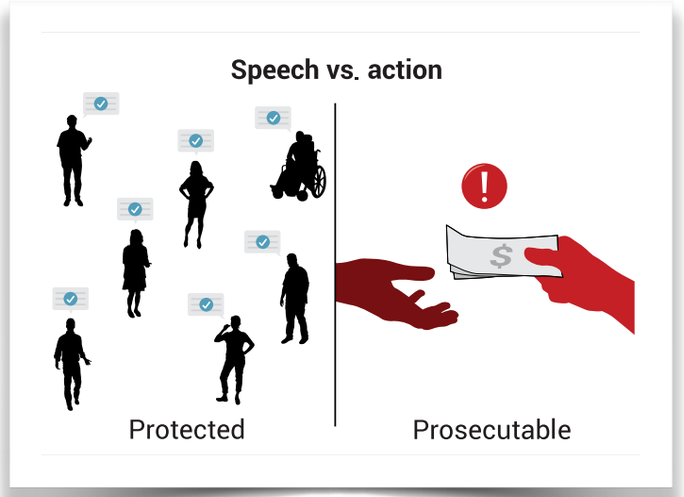
How do you solve a problem like **misinformation**?

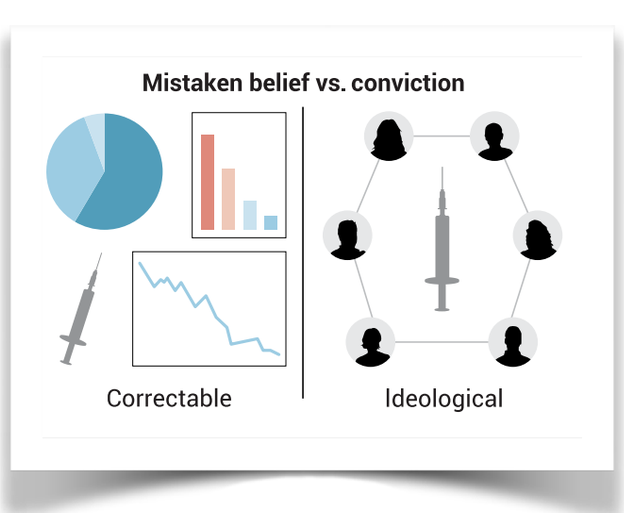
The pandemic was planned. Climate change is a hoax. Joe Biden lost the election.

Trying to navigate misinformation about COVID, climate change, politics, and countless other topics can be overwhelming. This is true for the public, researchers, journalists, and policy-makers alike. As researchers dedicated to the study and resistance of misinformation, we often find ourselves in conversation with government officials and others trying to understand and address the phenomenon. To help illuminate the complexities of misinformation and to guide policy, we find three distinctions helpful:

distinctions for research and policy







Calo *et al.*, *Sci. Adv.* **7**, eabn0481 (2021)

nature human behaviour

ARTICLES

https://doi.org/10.1038/s41562-022-01388-6

Check for updates

OPEN

Combining interventions to reduce the spread of viral misinformation

Joseph B. Bak-Coleman[1,2,3] ✉, Ian Kennedy[1,4], Morgan Wack[1,5], Andrew Beers[1,6], Joseph S. Schafer[1,6], Emma S. Spiro[1,3,4], Kate Starbird[1,7] and Jevin D. West[1,3]

Misinformation online poses a range of threats, from subverting democratic processes to undermining public health measures. Proposed solutions range from encouraging more selective sharing by individuals to removing false content and accounts that create or promote it. Here we provide a framework to evaluate interventions aimed at reducing viral misinformation online both in isolation and when used in combination. We begin by deriving a generative model of viral misinformation spread, inspired by research on infectious disease. By applying this model to a large corpus (10.5 million tweets) of misinformation events that occurred during the 2020 US election, we reveal that commonly proposed interventions are unlikely to be effective in isolation. However, our framework demonstrates that a combined approach can achieve a substantial reduction in the prevalence of misinformation. Our results highlight a practical path forward as misinformation online continues to threaten vaccination efforts, equity and democratic processes around the globe.

Stewardship of global collective behavior

Joseph B. Bak-Coleman[a,b,1], Mark Alfano[c,d], Wolfram Barfuss[e,f], Carl T. Bergstrom[g], Miguel A. Centeno[h], Iain D. Couzin[i,j,k], Jonathan F. Donges[e,l], Mirta Galesic[m], Andrew S. Gersick[n], Jennifer Jacquet[o], Albert B. Kao[m], Rachel E. Moran[a], Pawel Romanczuk[p], Daniel I. Rubenstein[n], Kaia J. Tombak[q], Jay J. Van Bavel[r,s], and Elke U. Weber[t,u]

Edited by Bonnie J. McCay, Rutgers, The State University of New Jersey, New Brunswick, NJ, and approved May 17, 2021 (received for review December 14, 2020)

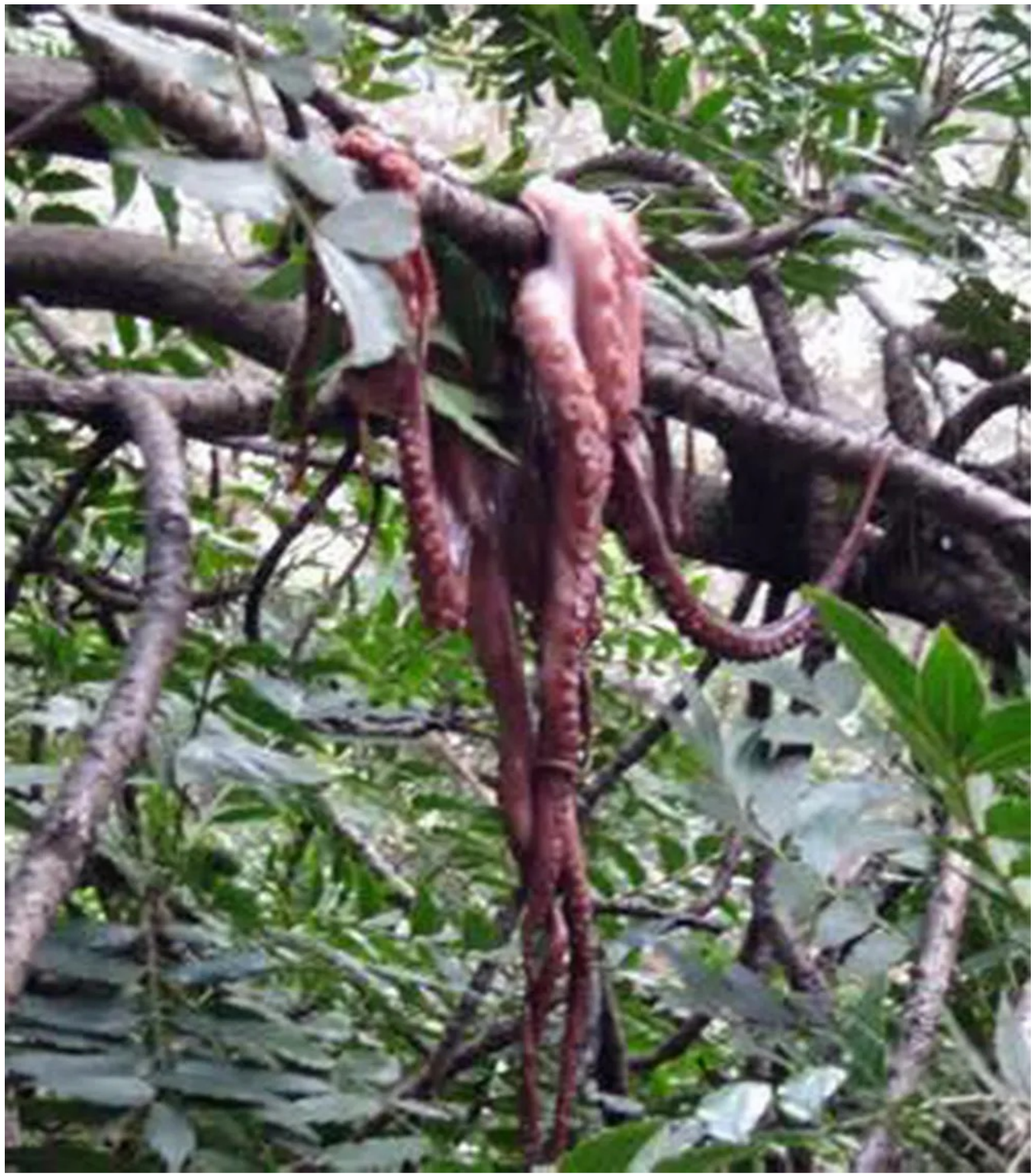
Collective behavior provides a framework for understanding how the actions and properties of groups emerge from the way individuals generate and share information. In humans, information flows were initially shaped by natural selection yet are increasingly structured by emerging communication technologies. Our larger, more complex social networks now transfer high-fidelity information over vast distances at low cost. The digital age and the rise of social media have accelerated changes to our social systems, with poorly understood functional consequences. This gap in our knowledge represents a principal challenge to scientific progress, democracy, and actions to address global crises. We argue that the study of collective behavior must rise to a "crisis discipline" just as medicine, conservation, and climate science have, with a focus on providing actionable insight to policymakers and regulators for the stewardship of social systems.

The coconut crab (*Birgus latro*) lives only on the tropical islands of Indian and Pacific Oceans



https://seasia.co/2019/06/08/hoooly-crab-meet-the-largest-land-arthropod

The Pacific Northwest tree octopus (paxarbolis)







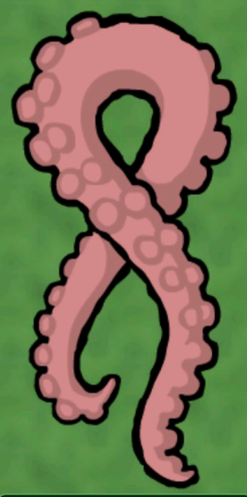
https://seasia.co/2019/06/08/endangered-olympic-penninsula-arboreal-cephalopod

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Help Save The ENDANGERED

PACIFIC NORTHWEST TREE OCTOPUS

From EXTINCTION!

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THE PACIFIC NORTHWEST TREE OCTOPUS



Rare photo of the elusive tree octopus
(Enhanced from cropped telephoto)

The Pacific Northwest tree octopus (*Octopus paxarbolis*) can be found in the **temperate rainforests** of the Olympic Peninsula on the west coast of North America. Their habitat lies on the Eastern side of the Olympic mountain range, adjacent to Hood Canal. These solitary cephalopods reach an average size (measured from arm-tip to mantle-tip,) of 30-33 cm. Unlike most other cephalopods, tree octopuses are amphibious, spending only their early life and the period of their mating season in their ancestral aquatic environment. Because of the moistness of the rainforests and specialized skin adaptations, they are able to keep from becoming desiccated for prolonged periods of time, but given the chance they would prefer resting in pooled water.

An intelligent and inquisitive being (it has the largest brain-to-body ratio for any mollusk), the tree octopus explores its arboreal world by both touch and sight. Adaptations its ancestors originally evolved in the three dimensional environment of the sea have been put to good use in the spatially complex maze of the **coniferous Olympic rainforests**. The challenges and richness of this environment (and the intimate way in which it interacts with it,) may account for the tree octopus's advanced behavioral development. (Some evolutionary theorists suppose that "arboreal adaptation" is what laid the groundwork in primates for the evolution of the human mind.)

Reaching out with one of her eight arms, each covered in sensitive suckers, a tree octopus might grab a branch to pull herself along in a form of locomotion called tentaculation; or she might be preparing to strike at an insect or small vertebrate, such as a frog or rodent, or steal an egg from a bird's nest; or she might even be examining some object that caught her fancy, instinctively desiring to manipulate it with her dexterous limbs (really deserving the title "sensory organs" more than mere "limbs",) in order to better know it.



Tree octopuses have eyesight comparable to humans. Besides allowing them to see their prey and environment, it helps them in inter-octopus relations. Although they are not social animals like us, they display to one-another their emotions through their ability to change the color of their skin: red indicates anger, white fear, while they normally maintain a mottled brown tone to blend in with the background.

The reproductive cycle of the tree octopus is still linked to

CEPHALONEWS

2022-08-17 **Another Path To Intelligence** (Nautilus)
Octopus brains are nothing like ours -- yet we have much in common.

2022-06-30 **Octopus Sightings Surge In Cornwall Waters** (BBC)
Cornwall Wildlife Trust said the last significant population boom for octopus was recorded on England's south coast in 1948. It is not known whether this year will classify as the first octopus boom in more than 70 years.

2022-06-24 **Italian Researchers A Step Closer To Understanding Cognitive Abilities Of The Octopus** (European Scientist)
Researchers found that some crucial genes -- called transposons or jumping genes -- are active both in the human brain and in the brains of some species of octopus. (Original paper: **"Identification of LINE retrotransposons and long non-coding RNAs expressed in the octopus brain"**)

» **Older news...**

More cephalonews at:

The Florida mangrove tree crab (Aratus pisonii)

